Examining The Factors That Affect Usage of UPI Transactions Across India

By

Krithika Kannan

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Submitted to Dr. Shubhashree Joshi, Department of Statistics

St. Joseph's University, Bangalore

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Abstract: This paper performs a state-wise examination of per capita UPI transactions in India. The paper aims to establish a connection between digital payments and other socio-economic and technological factors in a state, such as per capita income and access to banking services. Through this, the study hopes to establish digital payments as a marker of socio-economic and technological progress. Using data of 8 states in India, exploratory data analysis and regression analysis is done. Although the small sample restricts the accuracy of the results, connections are obtained with moderate confidence. Scope remains to perform further, more detailed study in this area.

Introduction:

UPI transactions and digital payments are a move towards a cashless economy in India. The RBI chief recently claimed that India accounts for 46% of digital transactions globally, with 80% of India's digital transactions being done through UPI. Digital payments provide several benefits over cash, including control over the black money market, increased convenience in making payments, ease of transaction, security and transparency. Widespread digital payments can be taken as a marker of technological and financial development. However, the adoption of digital payments varies across regions in India.

In addition to this, the popularity of UPI transactions in India has been growing rapidly. The value of UPI transactions done in India went from INR 6952.14 crore in FY2016-17 to INR 1,39,20,675 crore in FY2022-23. In June 2023 alone, the total value of UPI transactions was INR 14,75,464 crore. This immense volume can be comprehended on ground as well- in Bengaluru, even small carts and street vendors have QR codes by which people can make digital payments. The ease and speed of business seems to have greatly increased due to this technology.

In light of this, this paper seeks to find a connection between the number of UPI transactions in a state and factors of its Per Capita Income, number of MSMEs, cell phone network access, access to banking services, ownership of technological appliances such as mobile phone and laptop, and population. Correlation analysis will be done on each of these metrics to find out if any of these factors are linked to the levels of digital payments being made in these states. For this study, the chosen states/union territories are Karnataka, Andhra Pradesh, Gujarat, Maharashtra, Haryana, Uttar Pradesh, Assam and West Bengal.

Finally, a study will also be done examining the growth of some of these factors on a national scale.

Literature Review:

Ashish Baghla (2018) outlines the various kinds of digital transactions that can be done today. They establish that primary challenges in adoption of digital payments include lack of trust due to security issues, delay in cashback processing, loss of internet connectivity, and lack of knowledge and awareness among less educated people.

Sanghita Roy, Dr. Indrajit Sinha (2014) use the Technology Acceptance Model to examine the factors affecting adoption of digital payments. They found that the higher the current adoption of

digital payments, the faster it will grow. Perceived ease of use of the technology is the most significant factor affecting growth.
Tan and Teo (2000) establish that under the Diffusion of Innovation (DOI) theory, four additional constraints on adoption are relative advantage, compatibility, complexity and trialability.
Shruti Jose (2022) found that the main transition towards a cashless society started due to the result of the pandemic in India. They also found that the highest number of digital payments are done for shopping and recreational expenditure. The inability to make physical transactions during the pandemic caused an increase in digital payments in India.

Methodology:
The states/union territories for the study were selected using the data on the NPCI Digi Dhan dashboard- specific levels are mentioned in Table 1. Open-source data about the Per Capita Income of these states/union territories, the revenue share of their industries, the number of MSMEs in the regions, mobile towers installed in the regions by various TSPs (Telecommunication Service Providers) and the 2011 census data about access to banking services and ownership of appliances such as mobile phones was collected. Exploratory data analysis and regression analysis is done on these metrics to uncover links between UPI transactions and these factors. Tan and Teo (2000) establish a Diffusion of Innovation (DOI) Theory whereby adoption of innovations is linked to four constraints- relative advantage, compatibility, complexity and trialability. The factors examined in this paper were selected either as indicators of compatibility under the DOI theory or as markers of welfare and economic development.

Expected Outcome:
By examining the connections between the above-mentioned factors and the number of UPI transactions done in the mentioned states/union territories, this paper hopes to uncover correlations between these metrics. UPI has revolutionized the way that India makes payments and has increased ease of business. This paper hopes to quantify this as well as measure the increase in adoption of UPI payments against the other factors enumerated. Some of these factors indicate socio-economic development and technological advancement, while others indicate the general economic situation of the given region, and this paper hopes to connect the adoption of UPI payments as a marker of the progress in these areas.

Results:

The number of UPI transactions per capita for a state was collected for each of the states involved in the study. The data thus obtained was as follows:

State	UPI Transactions per Capita
Karnataka	4.994
Andhra Pradesh	17.005
Gujarat	1.844
West Bengal	1.558
Haryana	9.177
Uttar Pradesh	6.243
Maharashtra	3.605
Assam	0.727

Table 1: UPI transactions per capita per state

This can better be visualized using a pareto chart, to understand which states have lower and which states have higher number of transactions. This is done in chart 1:

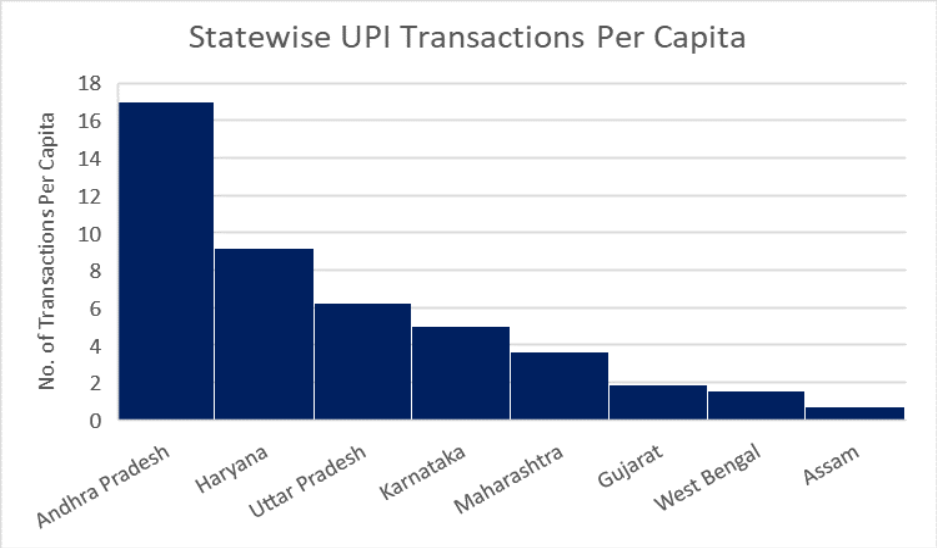

Chart 1: Number of UPI transactions per capita by state

The total UPI transactions by state can also be visualized in chart 2:

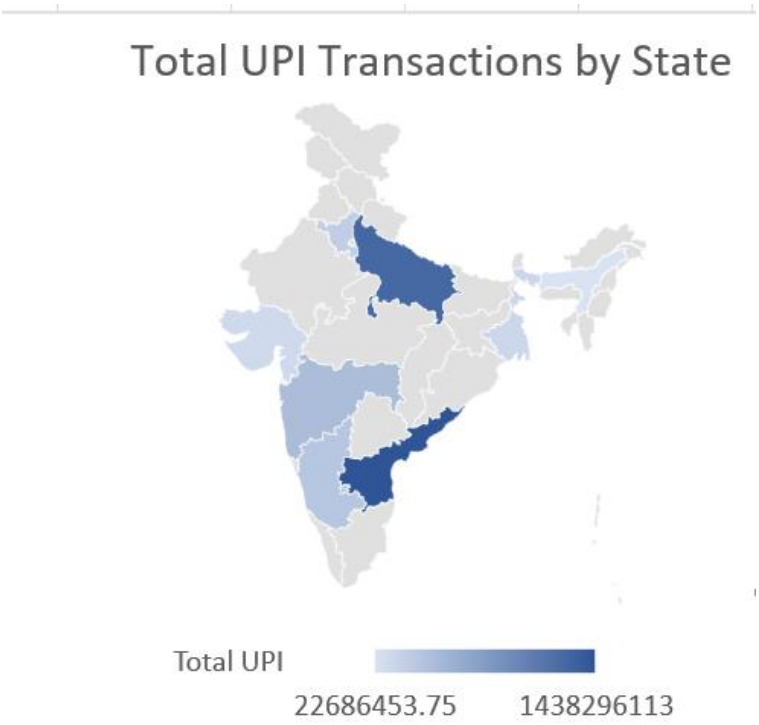

Chart 2: Map of total UPI transactions per state in the study

In this chart, the per capita UPI transactions are multiplied with the population of the states to get the total number of UPI transactions done in the state. We see that Andhra Pradesh and Uttar Pradesh have the highest numbers, followed by Maharashtra, Karnataka and Haryana, whereas in the pareto chart depicting per capita transactions, Andhra Pradesh, Haryana and Uttar Pradesh have the highest values, followed by Karnataka and Maharashtra. Clearly, the population density and size of the states makes the total transactions an inaccurate measure to compare against other standardized variables as the comparative magnitudes change, so using the per capita number of transactions will be better for analysis.

It is also seen that the southern states (Karnataka and Andhra Pradesh) along with the northern states (Uttar Pradesh and Haryana) have a higher number of transactions on average per capita than the eastern (West Bengal and Assam) and western (Gujarat and Maharashtra) states. The average UPI transactions per capita in the region is plotted against its standard deviation to obtain the following plot:

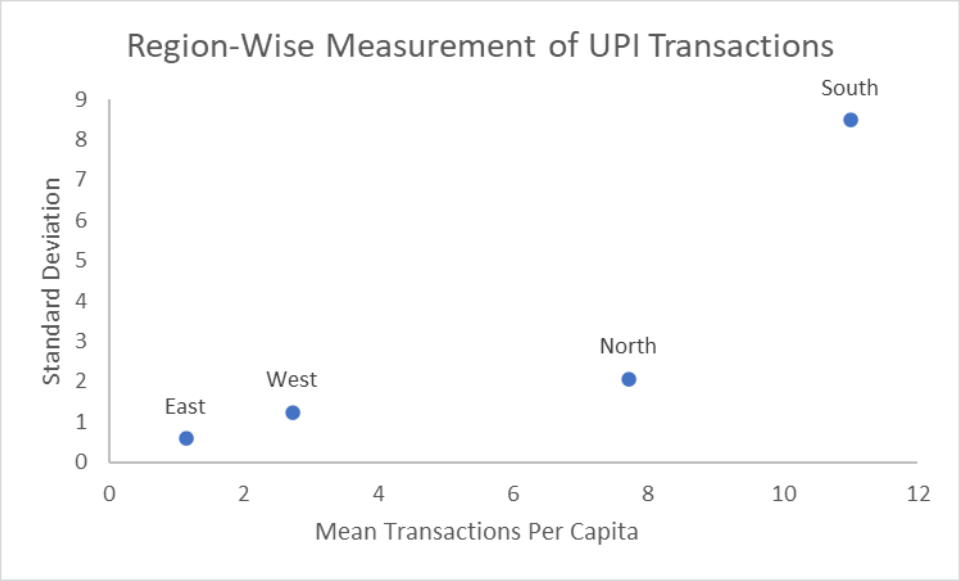

Chart 3: Plot of region-wise mean vs standard deviation of UPI transactions per capita

Here, we can see that the average UPI transactions per capita for the south is high with high variation, while in the north it is high with lower variation. In the east and west, the average number of UPI transactions per capita is uniformly quite low. Thus, we can say that there is some effect of the region of a state on the number of UPI transactions per capita done within it. The collected data is insufficient to conduct a proper correlation and regression analysis using the variables, but further study may yield fruitful results.

Furthermore, data about the population of the states in the study was collected. The relationship between population of the states and the number of UPI transactions can be viewed in the following scatter plot:

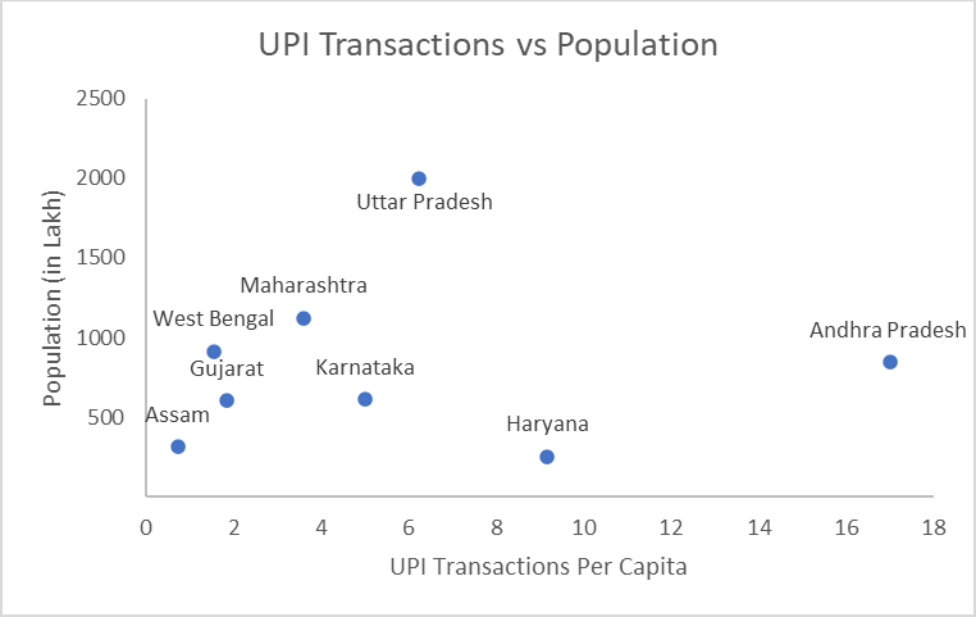

Chart 4: Relationship between UPI transactions and population of a state

The plot does not give us a very clear relationship, but there seems to be some level of positive correlation between the two variables. It also explains the change in comparative magnitudes of the total UPI transactions versus the per capita UPI transactions- Uttar Pradesh has a moderate per capita number of around 6 with high population density, Maharashtra and Karnataka have approximately 3 and 4 transactions per capita respectively and both have moderate populations. Haryana has a high number of transactions per capita but a very low population. These factors cause a re-ordering of the magnitudes when total transactions per state are considered.

After exploratory data analysis of the demographic variables, analysis of the economic factors of the study was done.
The per capita income of the states was collected and visualized as follows:

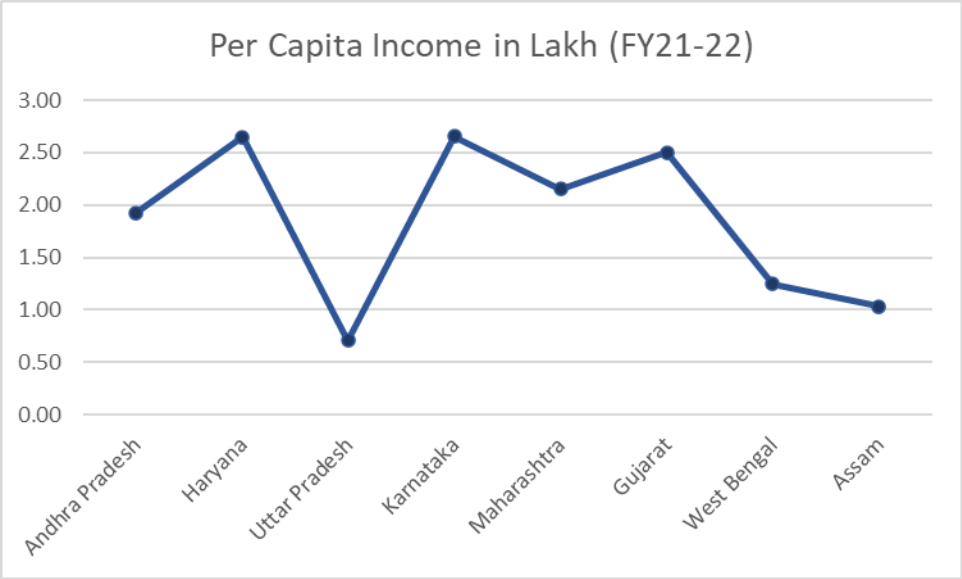

Chart 5: Per Capita Income by State in INR Lakh Per Annum

From an initial examination of chart 5 alone, we can tell that a clear relationship between UPI transactions and per capita income does not exist in the data being used. The states with the highest number of UPI transactions per capita (Andhra Pradesh, Haryana and Uttar Pradesh) all have highly varying per capita incomes. Karnataka, Maharashtra and Gujarat have similar incomes and there is a steep drop in per capita income for West Bengal and Assam, which at least is also consistent with the decrease in number of UPI transactions per capita in the eastern states.

To demonstrate this, the UPI transactions were then plotted against the per capita income of each state in the study. Upon plotting, it was found that no clear linear relationship exists between the two factors. These results can be viewed in chart 6:

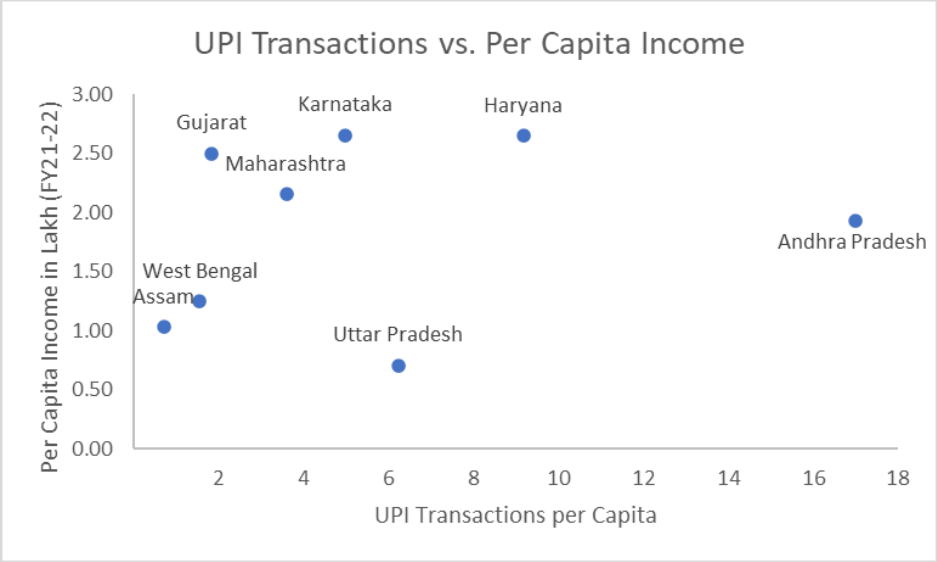

Chart 6: Relationship between UPI transactions and per capita income

We can see that in Andhra Pradesh, the per capita income is approx. INR 1.9 lakhs, and there are around 17 UPI transactions per capita. However, states with a higher per capita income such as Gujarat and Karnataka have lower numbers of UPI transactions. Similarly, Uttar Pradesh, with very low per capita income, has a higher number of UPI transactions than Maharashtra with a much higher per capita income. Thus, it is highly unlikely that these factors are correlated.

The second part of the analysis of economic factors was done regarding Micro, Small and Medium Enterprises (MSMEs). The number of MSMEs in each state was plotted in pareto charts to understand the data:

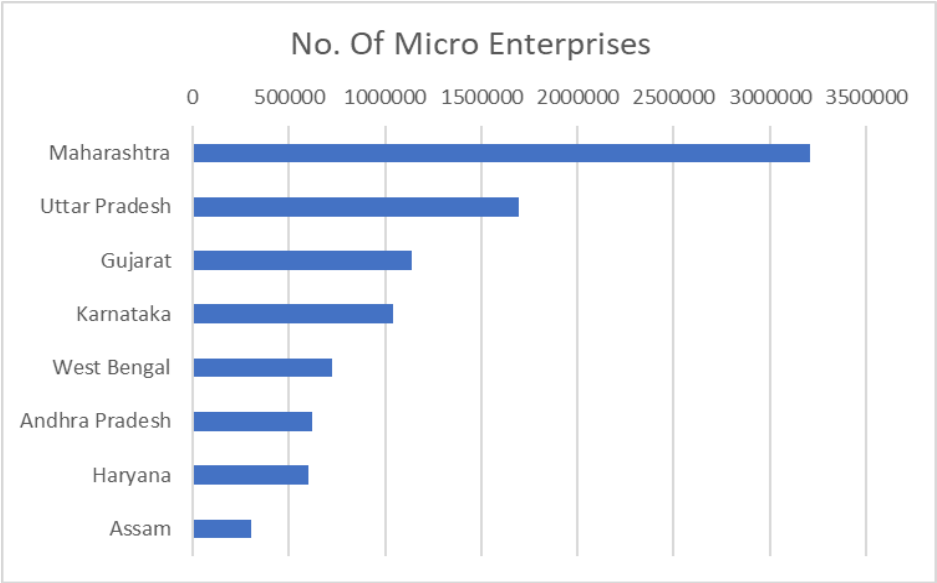

Chart 7: Micro Enterprises by State

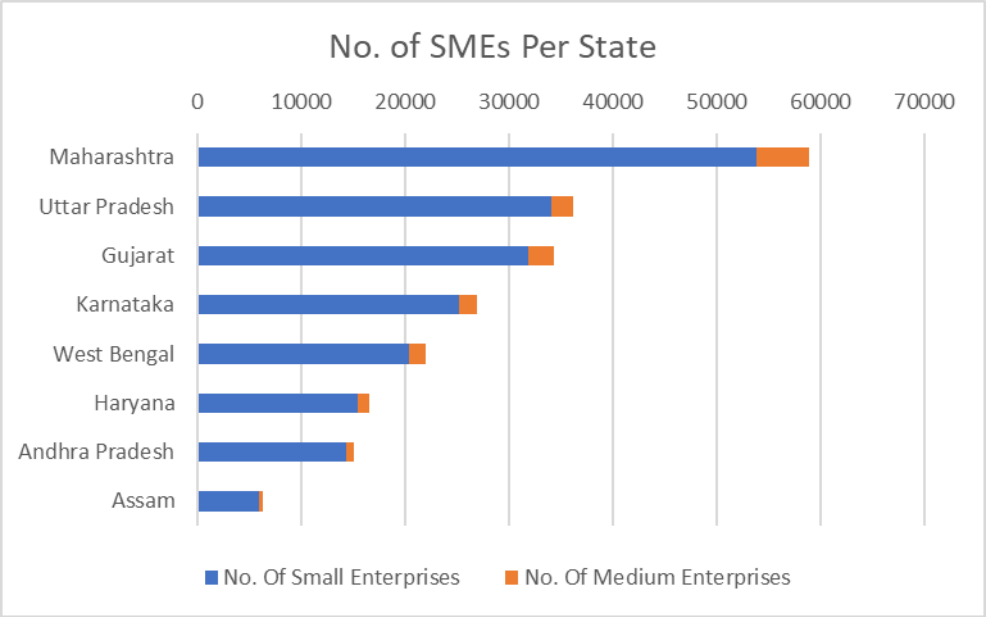

Chart 8: Small and Medium Enterprises by State

We can see from charts 7 and 8 that Maharashtra has the highest number of MSMEs, followed by Uttar Pradesh and Gujarat. Andhra Pradesh is closer to the bottom of the chart, with a comparatively low number of MSMEs, as is Haryana. Karnataka is in the middle of the pack. Once again, there is no clear correlation between the two variables. This can be better visualized using the following scatter plot:

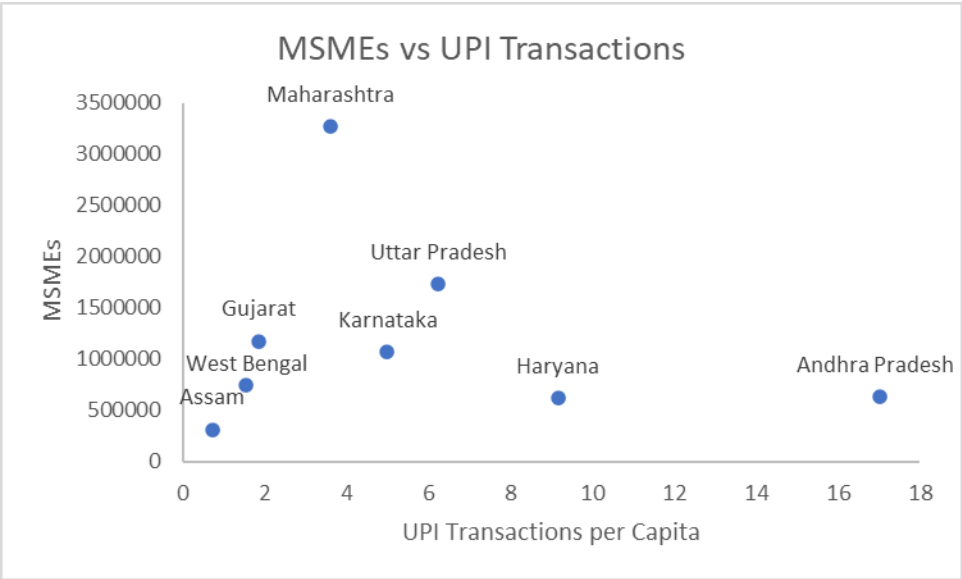

Chart 9: No. of MSMEs vs no. of UPI transactions by state

From the plot, we can conclude that perhaps a polynomial relationship exists between the variables, but a linear relationship, if it exists, is very weak.

After analysis of economic factors, data was collected regarding factors that ensure compatibility under the DOI theory. These factors are access to banking services and the number of wireless subscriptions (to various channels, such as BSNL, Reliance Jio and Airtel) in a state. Data about the mobile towers per square km installed by various Telecom Service Providers (TSPs) of the states was also collected. The distribution of these mobile towers among the various states can be visualized in chart 10.

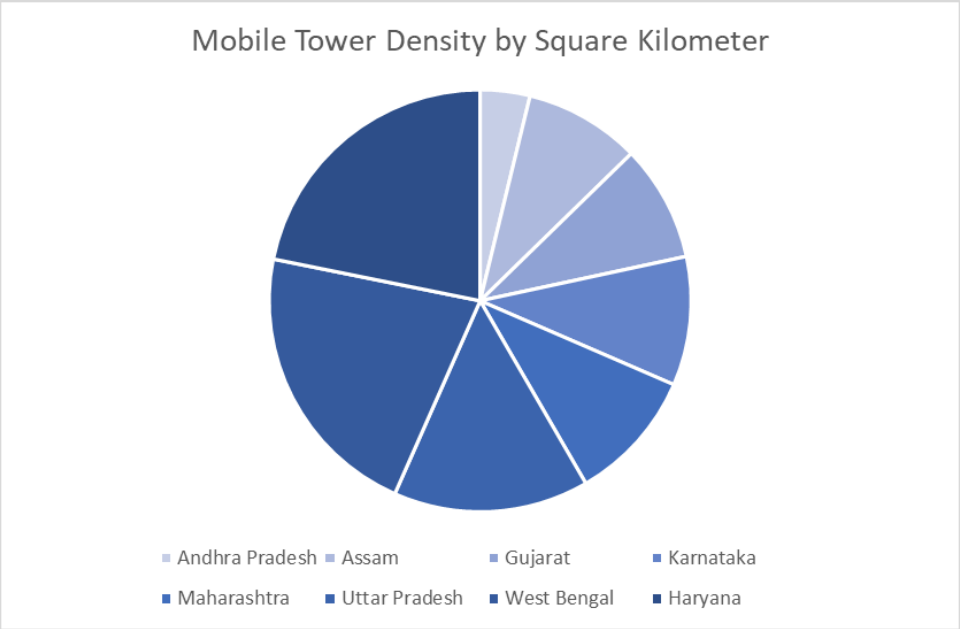

Chart 10: Mobile towers per square kilometer by state

In the plot, we see that Haryana and West Bengal have the highest number of mobile towers per square kilometer area. Andhra Pradesh surprisingly has the least density of mobile towers by square kilometer. Again, this indicates that there is no direct relationship whereby a higher or lower number of UPI transactions per capita in a state indicates that there may be a higher or lower number of mobile towers per square kilometer area in the state. However, this is the density for the overall state area. Most urban regions must have excellent connectivity and many mobile towers, and one can assume that urban regions are where the UPI transactions take place in larger numbers.

Upon plotting mobile towers in a state against the UPI transactions per capita, it was found that a weak linear relationship exists between the variables when some states are considered outliers. This can be viewed in the plot:

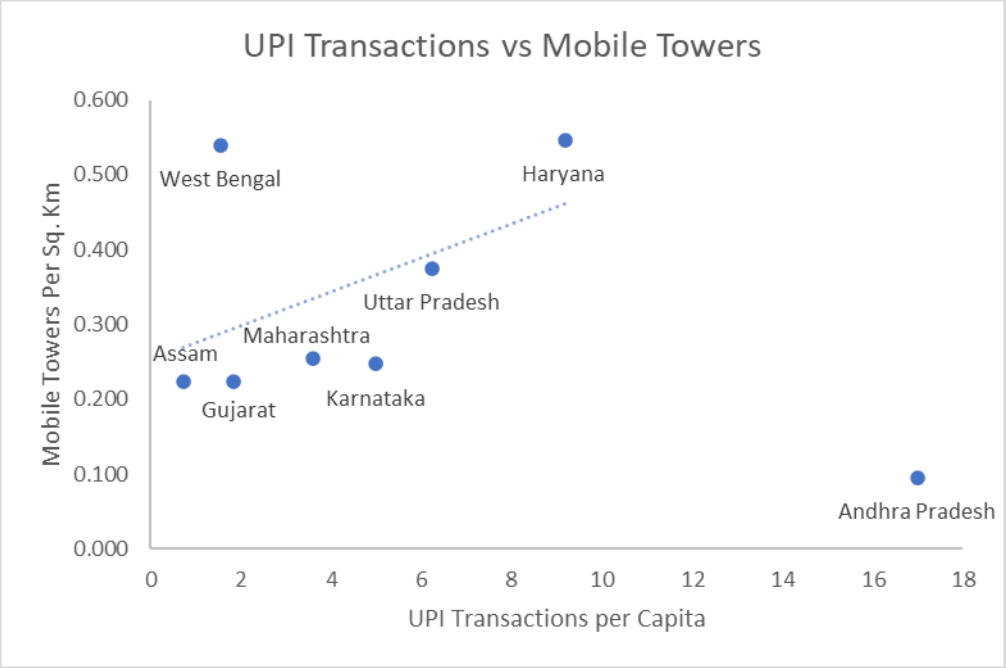

Chart 11: Mobile towers vs UPI transactions by state

The data about access to banking services can be plotted against UPI transactions to examine whether there is a relationship between the two variables. Intuitively, unless a person has access to a banking service, they cannot perform UPI transactions but it was desired to check whether the variables directly affect each other or not. Upon plotting, the relationship is unclear given the data at hand. This can be viewed in chart 12:

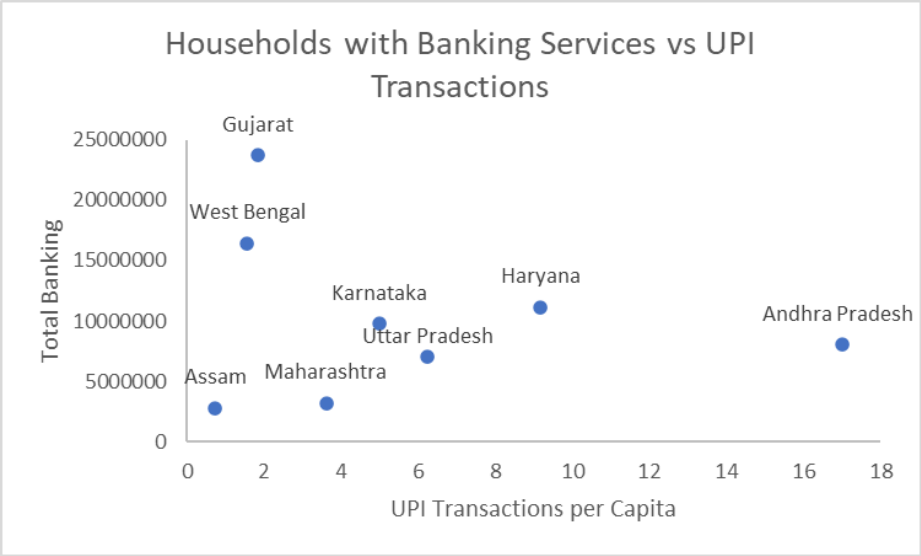

Chart 12: Plot of households with banking services against UPI transactions by state

To also understand how the number of households with access to banking services varies across states, we plot the values in a pie chart, as can be seen in chart 13:

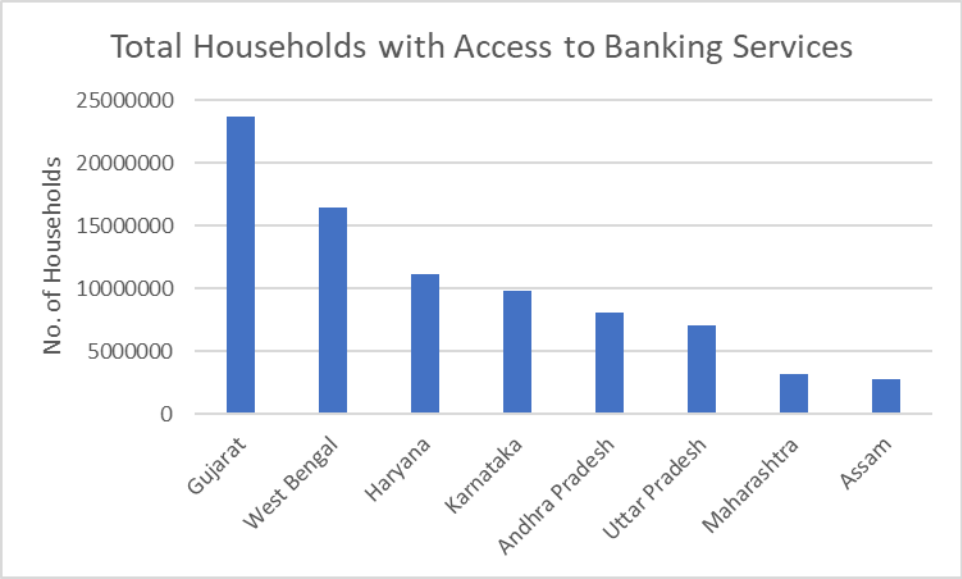

Chart 13: Households with Access to Banking Services by State

We see that Gujarat and West Bengal emerge at the top of the list. Haryana, Uttar Pradesh and Andhra Pradesh, the states with the highest per capita UPI transactions, have middling numbers when it comes to access to banking services. To accurately compare the two metrics, per capita UPI transactions were multiplied with the populations of the states to get the total UPI transactions done in the state. A pareto chart of the same is plotted in chart 14:

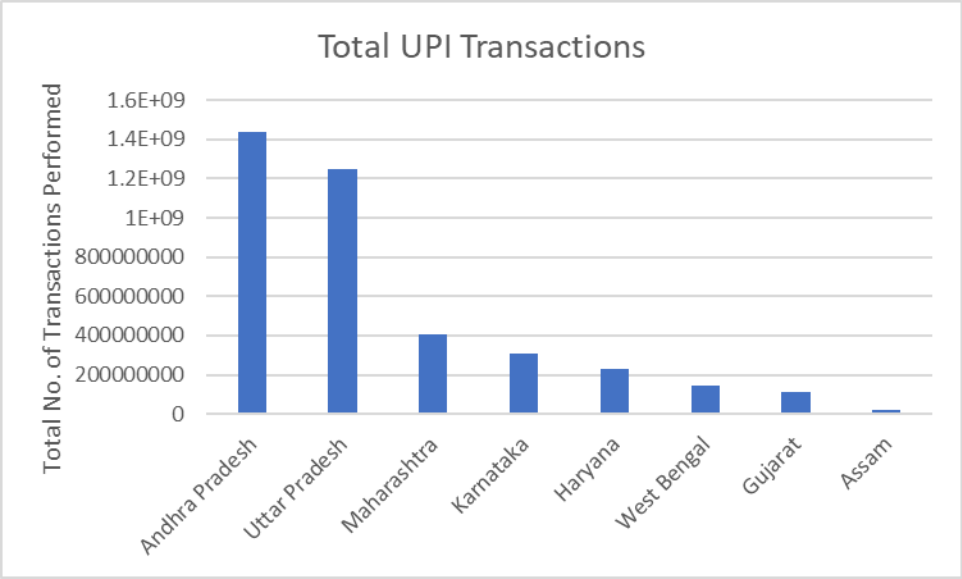

Chart 14: Total number of UPI transactions done per state

Here, we see that Andhra Pradesh and Uttar Pradesh have the highest total number of UPI transactions, whereas they have comparatively lower number of households with access to banking

services. Thus, there is no direct relationship between the two variables. While access to banking services is necessary for performing UPI transactions, other patterns such as average expenditures in the state and general trust of technology might be causing the relationship between the variables to be lost.

Finally, data about the connectivity of people in the state was measured through 3 variables- mobile towers per square kilometer area of the state, number of people in the state with a wireless subscription to some telecommunication service, and the over all 'tele-density' of the state, i.e., the standardized ratio of the level at which telecommunication services are used by residents of the state. The tele-density of the states in the study can be visualized in chart 15.

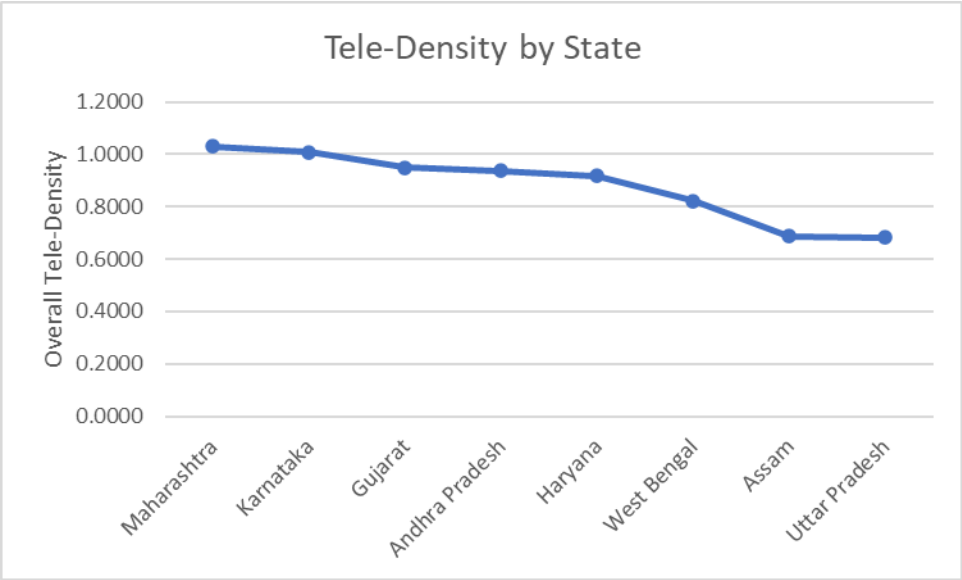

Chart 15: Tele-density by state (in percentage)

We find that Karnataka and Maharashtra have very high tele-density (over 100%), followed by Gujarat and Andhra Pradesh with close to 100% tele-density. This indicates that almost every person has access to telecommunication services, and some people have access to multiple forms of telecommunication service. While Assam and Uttar Pradesh have the lowest tele-density, all the states have high tele-density indicating that a fair share of the populations in these states has access to telecommunication services, thus implying that the compatibility for performing UPI transaction exists for them.

Upon plotting tele-density against number of UPI transactions per capita, it is found that a linear relationship can exist between the two variables. The Pearson's correlation coefficient was calculated to be 0.23, signifying either absence of or mild positive linear correlation between the variables. The plot can be examined in chart 16.

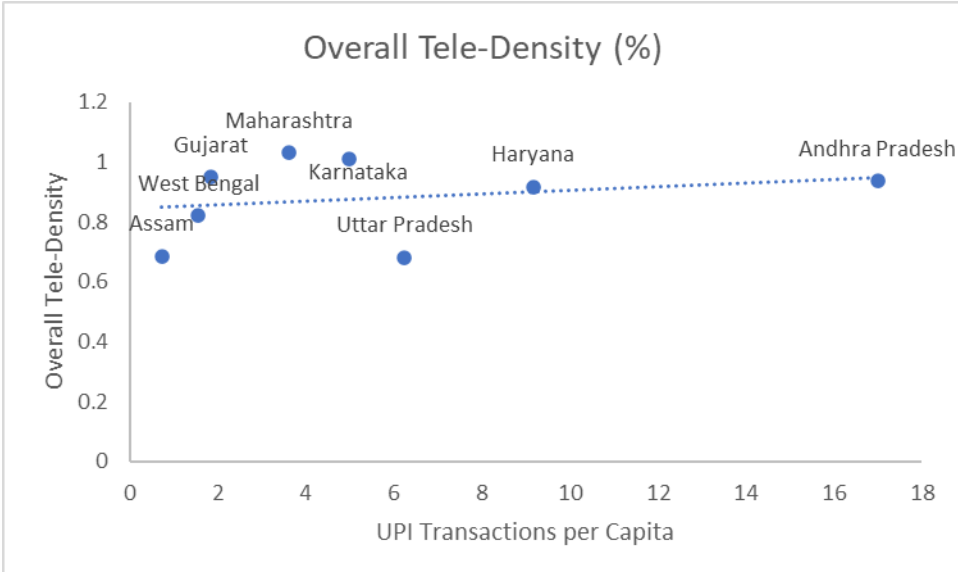

Chart 16: Tele-density vs UPI transactions per capita

Upon completion of the exploratory data analysis, it was clear that regression models using all the variables would not yield fruitful results, given the size of the sample and degree of variation in the data. An attempt was made to fit one linear model with total UPI transactions in a state as the dependent variable and urban population, households with banking services, and tele-density as the independent variables. This model was created in R, and the summary is as follows:

```
Call:
lm(formula = log(upidf$UPI * upidf$Population) ~ log(upidf$Urban) +
    log(upidf$Banking) + log(upidf$TeleDensity))

Residuals:
      1       2       3       4       5       6       7       8
-0.4329  0.9994 -0.2277  0.3091  0.7385  0.1288 -0.8349 -0.6803

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)              8.013      9.660   0.830   0.4534
log(upidf$Urban)        -2.744      1.751  -1.566   0.1923
log(upidf$Banking)       3.677      1.547   2.377   0.0762 .
log(upidf$TeleDensity)   8.014      4.578   1.750   0.1549
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.874 on 4 degrees of freedom
Multiple R-squared:  0.7584,    Adjusted R-squared:  0.5772
F-statistic: 4.186 on 3 and 4 DF,  p-value: 0.1002
```

We see that the multiple R-squared is 0.76. This indicates that the combination of factors results in moderate ability to predict the values of the dependent variable. The p-values for each independent variable are also not very high- at a significance level of 20% (i.e., 80% confidence), we can conclude that the variables can predict the value of the dependent variable.
The linear equation for predicting total UPI transactions (y) in a state based on the independent factors of urban population (u), number of households with banking services (h) and tele-density (d), is given by:

$$y = 8.013 - 2.744*u + 3.677*h + 8.014*d$$

The normal Q-Q plot was also done for the model, and the data is fairly normal, as can be seen in chart 17.

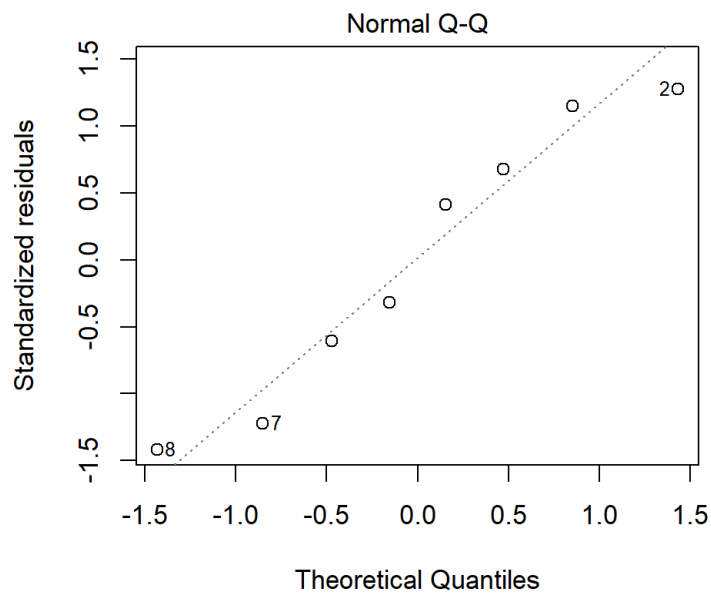

Chart 17: Normal Q-Q Plot for Linear Model

It must be noted that the regression analysis for other factors did not yield fruitful results due to the small sample size and high variation in the data. However, it can be noted that with a larger sample of richer data, either of all the states in India or even of all the districts, more accurate results can be obtained using more parameters.

Conclusion

While a very small sample of data was collected which resulted in the presence of outliers in the various comparisons, this indicates a high scope for further study in the area. With easily available data for all the states in India and, ideally, a district-wise breakdown of this data, both correlation and regression analysis would prove very fruitful and establish the growth and levels of digital payments usage in a state as a marker of its progress. Already, some relationship can be inferred

between UPI transactions and access to banking services, tele-density and urban population in a state. These factors are all good markers of development and upward mobility of social classes. Thus, we can establish with moderate confidence that the number of digital payments made in a state also serve as a marker of development.

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- https://data.gov.in/resource/stateut-wise-details-mobile-towers-installed-various-telecom-service-providers-tsps-across
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